

Program: Diploma in Engineering and Technology	
Course Code: 6421C	Course Title: Bakery and Confectionery
Semester: 6	Credits: 4
Course Category: Program Elective	
Periods per week: 4(L: 4T:0 P: 0)	Periods per semester: 60

Course Objectives:

1. To understand the fundamental raw materials required for a bakery and confectionary unit
2. To study the basic equipments and concepts required for setting up a bakery unit
3. To impart knowledge on various types of confectionaries and their processing

Course Prerequisites:

Topic	Program/Course Name
Basic knowledge of bakery and confectionery	Bakery and confectioneries

Course Outcomes:

On completion of the course, the student will be able to:

CO _n	Description	Duration (Hours)	Cognitive Level
CO1	Analyze the fundamental raw materials vital in bakery units	14	Understanding
CO2	Explain the important unit operations used in bakery and confectionery technology	16	Understanding
CO3	Explain the essential equipment, regulations, packaging requirements, and layout considerations for a bakery unit.	15	Understanding
CO4	Explain the types, classification, and processing methods of crystallized confectionery, liquorice, and gums	15	Understanding

CO–PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	2						
CO2	2						
CO3	2						
CO4	2						

3-Strongly mapped,2-Moderately mapped,1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Analyze the fundamental raw materials vital in bakery units		
M1.01	Summarize the history of bakery and confectionery	4	Understanding
M1.02	Discuss the raw materials used in Bakery	5	Understanding
M1.03	Describe the characteristic of flour types	5	Understanding
Contents: Introduction: History of bakery and confectionery-trends, Raw materials used in Bakery, Flour -types and characteristics			
CO2	Explain the important unit operations used in bakery and confectionery technology		
M2.01	Discuss various unit operations in a bakery unit	5	Understanding
M2.02	Classify the different types of breads	5	Understanding
M2.03	Explain the processing of Pizza and pastry	5	Understanding
	Series Test –I	1	
Contents: Unit operations in a bakery unit- Mixing-fermentation-proofing-Baking, Types of breads, Processing of Pizza and pastry			
CO3	Explain the essential equipment, regulations, packaging requirements, and layout considerations for a bakery unit.		

M3.01	Discuss the equipments required in a bakery unit	5	Understanding
M3.02	Summarize the regulations to be followed	5	Understanding
M3.03	Review the packaging requirements for bakery products, Bakery unit layout	5	Understanding
Contents: Equipment required in a bakery unit, Regulations to be followed-quality and standards, Packaging requirement for bakery products, Bakery unit layout			
CO4	Explain the types, classification, and processing methods of crystallized confectionery, liquorice, and gums		
M4.01	Classify types of confectioneries	3	Understanding
M4.02	Explain the processing of crystallized confectionary	4	Understanding
M4.03	Explain the processing of liquorice	4	Understanding
M4.04	Explain the processing of gums	3	Understanding
	Series Test–II	1	
Contents: Types of confectioneries, classification, Processing of crystallized confectionary Processing of liquorice, Processing of gums- chewing gum, bubble gum			

Text / Reference

T/R	Book Title/Author
R1	Bernard. W. Minifie., PhD “Chocolate, Cocoa, and confectionery” (Science and Technology), 3rd edition, CBS publishers and Distributors, New Delhi-110002
R2	Ewald Notter, The art of confectioner, sugar work and pastillage, Wiley Publishers
R3	Cauvain, Stanley P, and Young, Linda S., Technology of Bread Making, Second Edition Aspen publication. Maryland, 1999
R4	Matz, Samuel A., Bakery Technology and Engineering, Third Edition, Chapman & Hall, London.
R5	Y H Hui, Bakery products science and technology, Blackwell Publishers

Online Resources

Sl.No	Website Link
1	https://www.bakeinfo.co.nz/facts/bread/science-of-bread-making/
2	https://www.adinath.co.in/chewing-gum-and-bubble-gum-production-process-in-confectionary-industry/
3	https://opentextbc.ca/modernpastryandplateddesserts/chapter/crystallization/